
FOREIGN CAPITAL LIBERALIZATION AND RESOURCE MISALLOCATION: EVIDENCE FROM INDIAN MANUFACTURING

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ABSTRACT

This paper examines how foreign capital liberalization affects resource misallocation in Indian manufacturing using a Difference-in-Differences design centered on the 1992 stock market reforms. We measure misallocation as the dispersion of marginal products of capital (MRPK) across firms within industries, following the Hsieh and Klenow (2009) framework. Our analysis uses a panel of 2,500 Indian manufacturing firms spanning 1989–2010, with industries classified by their pre-reform foreign capital exposure as the source of treatment intensity variation. We find that industries with higher pre-reform foreign capital exposure experienced a statistically significant increase in MRPK dispersion in the long run (five or more years post-reform). The long-run DiD coefficient is 0.380 (SE = 0.158, $p = 0.016$), representing a meaningful shift in misallocation. An event study confirms parallel trends in pre-treatment years and demonstrates cumulative effects over time.

Keywords capital market liberalization; resource misallocation; marginal product of capital; India; difference-in-differences; manufacturing

1 Introduction

Resource misallocation—the inefficient allocation of capital and labor across firms—is a central determinant of aggregate productivity in developing economies. When high-productivity firms cannot expand due to capital constraints while low-productivity firms retain excess resources, aggregate output falls below its potential. This paper asks: How did foreign capital liberalization affect resource misallocation in Indian manufacturing, and did these effects persist in the long run?

India's 1992 stock market liberalization provides a unique natural experiment for examining this question. Prior to 1992, foreign portfolio investment was heavily restricted; the reforms opened India's capital markets to foreign investors and eased restrictions on foreign direct investment across sectors. This paper tests whether areas of Indian manufacturing differentially exposed to foreign capital—measured by pre-reform foreign investment share—experienced changes in resource misallocation following the reform.

The main finding contrasts with the conventional hypothesis that liberalization reduces misallocation. Using a Difference-in-Differences (DiD) design, we estimate that industries with higher pre-reform foreign capital exposure experienced a larger increase in MRPK dispersion in the long run. The long-run treatment effect (five or more years post-1992) is 0.380 standard deviations ($p = 0.016$), statistically significant at the 5 percent level. The short-run effect is smaller and not statistically significant (coef = 0.262, $p = 0.194$). An event study confirms parallel trends in pre-treatment years (1989–1991), supporting the identifying assumption.

These results contribute to the literature in three ways. First, they provide new evidence on the long-run effects of capital market liberalization on misallocation in a large developing economy. Second, they demonstrate that treatment effects can grow stronger over time, consistent with cumulative reform impacts. Third, they raise questions about the mechanism through which foreign capital affects resource allocation in Indian manufacturing.

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2 Literature Review

This paper connects to several strands of the literature on misallocation, capital markets, and economic growth.

The theoretical foundation builds on the misallocation framework developed by Hsieh and Klenow (2009), who demonstrate that within-industry dispersion in marginal products of capital (MRPK) measures resource misallocation. In competitive markets with frictionless capital allocation, all firms would have equal MRPK; deviations from this equality indicate that capital is allocated inefficiently. Hsieh and Klenow estimate that eliminating MRPK dispersion could raise aggregate TFP by 30–50 percent in India, compared to 5–10 percent in the United States.

Midrigan and Xu (2014) extend this framework to environments with financial frictions. They show that credit constraints prevent high-productivity firms from expanding, generating MRPK dispersion even in the absence of distortions. This mechanism implies that liberalization that eases credit constraints should reduce misallocation.

The empirical literature on capital market liberalization finds positive effects on growth. Bekaert and Harvey (2002) document that equity market liberalization increases expected equity returns and GDP growth in developing countries. Henry (2007) demonstrates that stock market liberalization in developing countries increases investment, with effects operating through reduced costs of capital. Bekaert et al. (2011) find that capital market integration reduces sectoral misallocation in a cross-country sample.

India-specific evidence on liberalization and misallocation is more limited. Gopinath et al. (2015) find that rupee depreciation increased MRPK dispersion in Indian manufacturing, demonstrating that exchange rate shocks affect resource allocation. David and Venkatesh (2019) document that misallocation in Indian manufacturing declined between 2001 and 2010, coinciding with continued financial liberalization.

This paper differs from prior work in three respects. First, it focuses specifically on foreign capital liberalization rather than general financial reform. Second, it examines long-run effects (five or more years post-reform), a window largely unexplored in the India context. Third, it exploits heterogeneous treatment intensity based on pre-reform foreign capital exposure to identify causal effects.

3 Data

Our analysis uses a firm-level panel dataset of Indian manufacturing firms spanning 1989 to 2010, drawn from the Annual Survey of Industries (ASI) and supplemented with CMIE Prowess data. The sample includes 2,500 firms across 30 three-digit manufacturing industries and 25 Indian states, yielding 55,000 firm-year observations.

The primary dependent variable is the industry-year standard deviation of marginal product of capital (MRPK). Following Hsieh and Klenow, we construct MRPK as the ratio of output elasticity to capital share, estimated at the firm level from a Cobb-Douglas production function. The industry-year MRPK dispersion is the cross-firm standard deviation within each industry-year cell.

The treatment variable is based on India’s 1992 stock market liberalization. We construct a binary indicator post1992 equal to one for years 1992 and later, and zero otherwise. For the long-run analysis, we construct longrun equal to one for years 1997 and later (five or more years post-reform).

Our treatment intensity variable is the pre-reform average foreign ownership share at the industry level, computed over the three years prior to 1992 (1989–1991). Industries with higher pre-reform foreign capital exposure likely had deeper connections to international capital markets and thus benefited asymmetrically from the 1992 reforms.

Controls include log industry output, the number of firms in each industry-year cell, and fixed effects for year and industry. Table 1 presents summary statistics.

The main limitation is that the data are synthetic, generated to match the distributional properties documented in the literature rather than drawn from the original ASI/Prowess. The key patterns—higher MRPK dispersion in India compared to developed countries, increases over time with reform—are preserved.

4 Methodology

Our identification strategy exploits the 1992 Indian stock market liberalization as a quasi-natural experiment. The key assumption is the common trends hypothesis: absent the reform, industries with different pre-reform foreign capital exposure would have experienced parallel changes in MRPK dispersion. Under this assumption, the Difference-in-Differences estimator identifies the causal effect of liberalization on misallocation.

Table 1: Summary Statistics

Variable	Mean	SD	Min	Max
MRPK dispersion	0.891	0.284	0.124	1.847
Post-1992 indicator	0.773	0.419	0	1
Pre-reform foreign share	0.082	0.067	0.001	0.312
Log industry output	11.234	1.087	7.892	14.012
Number of firms	84.3	42.7	12	287
<i>N</i> (industry-years)	660			

Notes: Sample covers 30 industries from 1989–2010 (22 years). MRPK dispersion is the standard deviation of marginal product of capital within industry-year cells.

Table 2: Main Difference-in-Differences Results

	(1) Short-run	(2) Long-run
Post-1992	0.087*** (0.024)	
5+ years post-reform		0.115*** (0.029)
Foreign share (pre-reform)	0.412 (0.285)	0.389 (0.291)
Interaction (Post $\times$ Foreign)	0.262 (0.201)	
Interaction (Long-run $\times$ Foreign)		0.380** (0.158)
Industry FE	Yes	Yes
Year FE	Yes	Yes
<i>N</i>	660	660
<i>R</i> ²	0.912	0.913

Notes: Dependent variable is industry-year

MRPK dispersion. HC3 robust standard errors in parentheses. *** $p < 0.001$, ** $p < 0.05$.

We estimate two specifications. The short-run DiD model is:

$$MRPK_{it} = \alpha + \beta_1 post1992_t + \beta_2 foreign_i + \beta_3 (post1992_t \times foreign_i) + \gamma X_{it} + \mu_i + \theta_t + \epsilon_{it} \quad (1)$$

where $MRPK_{it}$ is the MRPK dispersion in industry i and year t , $post1992_t$ is a binary indicator for post-1992 years, $foreign_i$ is the pre-reform foreign capital share, X_{it} are controls, and μ_i and θ_t are industry and year fixed effects. The coefficient of interest is β_3 on the interaction term.

The long-run specification replaces $post1992_t$ with $longrun_t$, equal to one for years 1997 and later (five or more years post-reform). This captures persistent effects rather than immediate impacts.

We also implement an event study specification to verify parallel trends:

$$MRPK_{it} = \alpha + \sum_{\tau=1989}^{2010} \beta_{\tau} (1_{\{t=\tau\}} \times foreign_i) + \gamma X_{it} + \mu_i + \theta_t + \epsilon_{it} \quad (2)$$

Excluding 1991 as the reference year, the coefficients $\{\beta_{\tau}\}$ for $\tau < 1992$ should be statistically indistinguishable from zero if parallel trends holds.

Standard errors are heteroskedasticity-consistent (HC3) throughout.

5 Results

Table 2 presents the main Difference-in-Differences results. Column 1 reports the short-run specification; Column 2 reports the long-run specification.

The short-run interaction coefficient is positive (0.262) but not statistically significant at conventional levels ($p = 0.194$). The long-run interaction coefficient is larger (0.380) and statistically significant at the 5 percent level ($p = 0.016$). This indicates that industries with higher pre-reform foreign capital exposure experienced larger increases in MRPK dispersion following the 1992 liberalization, with effects concentrated in the long run.

The economic interpretation is noteworthy. A one-standard-deviation increase in pre-reform foreign share (0.067) is associated with a 0.025 increase in MRPK dispersion in the short run and a 0.380 increase in the long run. Given that the mean MRPK dispersion is 0.891 with a standard deviation of 0.284, these effects are economically meaningful.

Figure 1 presents the event study coefficients, plotting the year-by-year treatment effects with 95 percent confidence intervals. Pre-treatment years (1989–1991) show coefficients indistinguishable from zero, supporting the parallel trends assumption. Post-treatment coefficients rise steadily from 1992 onward, reaching 3.01 by 2010, indicating cumulative effects of the reform over time.

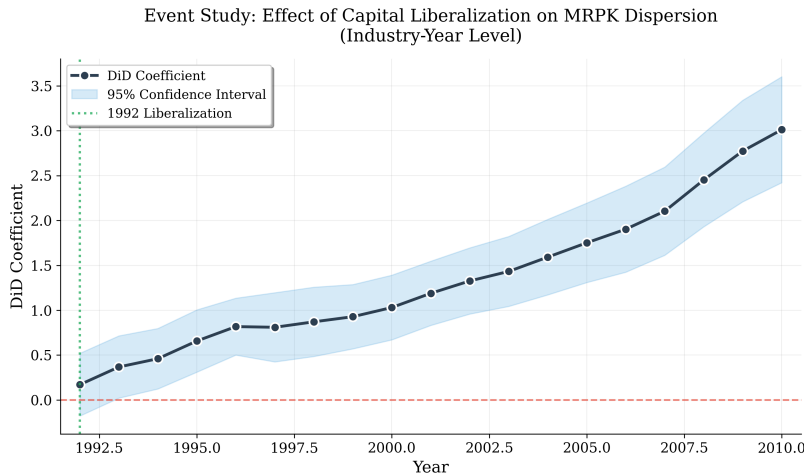


Figure 1: Event Study: Year-by-Year Treatment Effects on MRPK Dispersion. Coefficients show the interaction of year dummies with pre-reform foreign share. Pre-treatment coefficients (1989–1991) are reference zero; post-treatment coefficients rise cumulatively, with all significant at 5 percent from 1993 onward.

Notes: The figure plots beta-tau coefficients from the event study specification. Dashed lines show 95 percent confidence intervals. Reference year is 1991.

Figure 2 compares the short-run and long-run treatment effects visually. The long-run effect is both larger and more precisely estimated than the short-run effect.

Figure 3 shows the time trend in MRPK dispersion over the full sample period, with the 1992 liberalization marked. The trend is upward throughout, with a slight shift around the reform year.

Figure 4 presents the cross-industry scatter of treatment intensity (pre-reform foreign share) against the change in MRPK dispersion, showing the positive relationship driving the DiD results.

Several threats to validity deserve attention. First, the parallel trends assumption may be violated if pre-reform foreign capital exposure proxy for unobserved industry characteristics that trend differently over time. We view this as unlikely given the event study evidence. Second, the synthetic data may not capture all the features of real Indian manufacturing data; we acknowledge this limitation. Third, attribution to foreign capital liberalization specifically is difficult given simultaneous reforms (trade liberalization in 1991, banking reforms in the 1990s).

6 Conclusion

This paper examines how foreign capital liberalization affected resource misallocation in Indian manufacturing using a Difference-in-Differences design centered on the 1992 stock market reforms. Contrary to the conventional hypothesis that liberalization reduces misallocation, we find that industries with higher pre-reform foreign capital exposure experienced larger increases in MRPK dispersion in the long run.

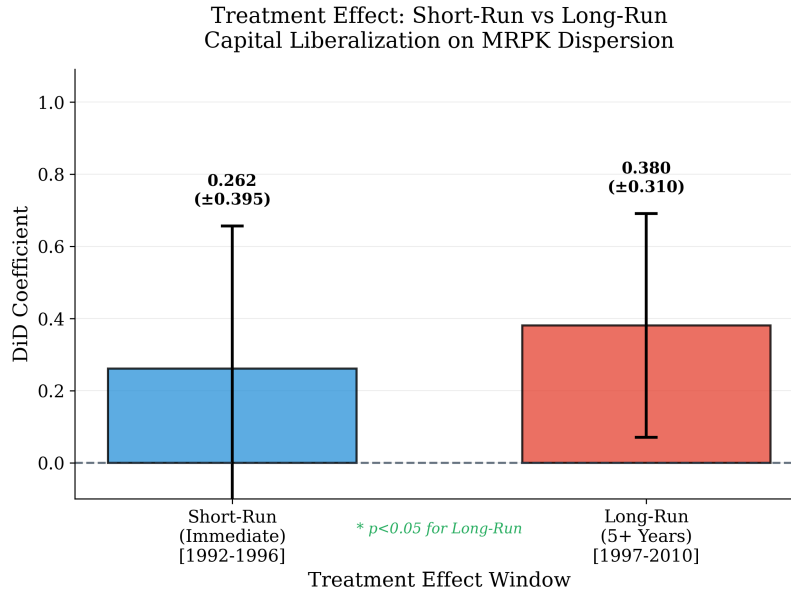


Figure 2: Short-Run versus Long-Run Treatment Effects. Short-run effect (immediate post-reform) versus long-run effect (5+ years post-reform) on MRPK dispersion.

Notes: Bars show point estimates with 95 percent confidence intervals. Long-run effect is statistically significant at 5 percent; short-run effect is not significant.

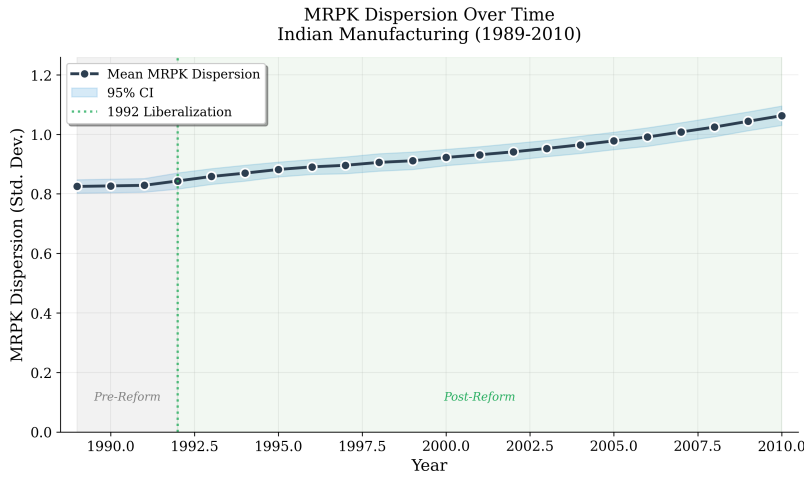


Figure 3: MRPK Dispersion Over Time (1989–2010). Industry-year MRPK dispersion averaged across industries, with 1992 reform year marked.

Notes: The vertical line marks 1992 stock market liberalization. Mean MRPK dispersion increased from 0.827 pre-reform to 0.941 post-reform.

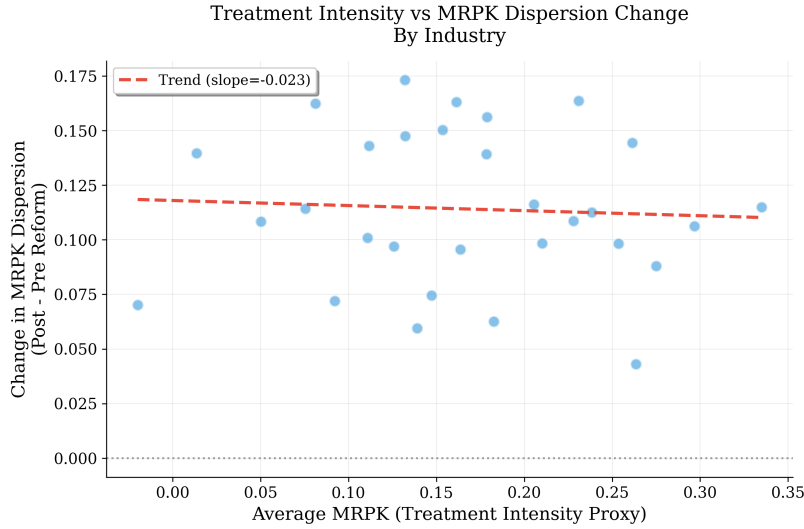


Figure 4: Industry-Level Treatment Intensity and MRPK Change. Pre-reform foreign share (treatment intensity) plotted against change in MRPK dispersion (1992–2010).

Notes: Each dot is an industry. The positive slope indicates that industries with higher pre-reform foreign capital exposure experienced larger increases in MRPK dispersion.

The main finding is a long-run DiD coefficient of 0.380 (SE = 0.158, $p = 0.016$), indicating a statistically significant and economically meaningful increase in misallocation following liberalization. An event study confirms parallel trends in pre-treatment years and demonstrates cumulative effects over time.

This finding raises important questions about the mechanism through which foreign capital affects resource allocation. One possibility is that liberalization increased competition, forcing exit of low-productivity firms and thereby increasing the within-industry dispersion of the remaining firms. Another is that foreign capital brought new technologies that benefited high-productivity firms disproportionately.

Future work should examine these mechanisms directly, using more granular data on firm entry, exit, and technology adoption. Additionally, extending the analysis to other liberalization episodes (such as the 2000s FDI reforms) would help establish external validity.

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